

Problem 6.7

- (a) Calculate the expectation value $\langle r \rangle_{21}$ for the hydrogen atom and compare it with the value r at which the radial probability density reaches its maximum for the state $n = 2, l = 1$.
(b) Calculate the width of the probability density distribution for r .

① $\psi_{n\ell m} = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$

Recall:

$$\begin{aligned}\langle r \rangle &= \langle n, \ell, m | \hat{r} | n, \ell, m \rangle \\ &= \iiint r |\psi_{n\ell m}|^2 r^2 \sin\theta \, dr d\theta d\phi \\ &= \left(\int_0^\infty r |R_{n\ell}|^2 r^3 dr \right) \underbrace{\left(\int_0^{2\pi} d\phi \int_0^\pi \sin\theta |Y_{\ell m}|^2 d\theta \right)}_{=1} \\ &= \int_0^\infty r^3 |R_{n\ell}|^2 dr\end{aligned}$$

Now, the notation:

$$\langle r \rangle_{21} = \int_0^\infty r^3 |R_{21}(r)|^2 dr$$

Read this off of the table

$$R_{21}(r) = \frac{1}{\sqrt{24}a_0^{5/2}} r e^{-r/2a_0}$$

$$\begin{aligned}\langle r \rangle_{21} &= \int_0^\infty r^3 \left| \frac{1}{\sqrt{24}a_0^{5/2}} r e^{-r/2a_0} \right|^2 dr \\ &= \frac{1}{24a_0^5} \int_0^\infty r^5 e^{-r/a_0} dr\end{aligned}$$

• let $u = r/a_0$

$$du = dr/a_0 \Rightarrow dr = a_0 du$$

$$\begin{aligned}\langle r \rangle_{21} &= \frac{1}{24a_0^5} \int_0^\infty (a_0 u)^5 e^{-u} (a_0 du) \\ &= \frac{1}{24a_0^5} \cdot a_0^6 \left(\int_0^\infty u^5 e^{-u} du \right) \\ &= \frac{a_0}{24} \cdot (5!) = 5a_0\end{aligned}$$

or

$$\langle r \rangle_{21} = 5a_0$$

→ why is this $= 5!$? Note that the gamma-function takes the form

$$\Gamma(n+1) = n! = \int_0^\infty x^n e^{-x} dx$$

where n is a positive integer

$$n = 0, 1, 2, 3, \dots$$

we see that our integral has this form

$$\int_0^\infty x^5 e^{-x} dx$$

so $5 = n$ and so

$$\int_0^\infty x^5 e^{-x} dx = 5!$$

- Compare with (a) part (ii) of previous problem to solve second half.

(b) Width of probability density: (Review)

- Recall that $\langle r \rangle$ gives us the average value of r .

- If we measure some distance from this average:

$$\delta = r - \langle r \rangle$$

This implies

$$\langle \delta \rangle = \langle r - \langle r \rangle \rangle = \langle r \rangle - \langle r \rangle = 0$$

which says the mean value of the deviation from $\langle r \rangle$ is zero.

- To get a non-zero value we square this term

$$\delta^2 = (r - \langle r \rangle)^2$$

$$\delta^2 = r^2 - 2r\langle r \rangle + \langle r \rangle^2$$

we then take the average of this term

$$\begin{aligned}\langle \delta^2 \rangle &= \langle r^2 \rangle - 2\langle r \rangle^2 + \langle r \rangle^2 \\ &= \langle r^2 \rangle - \langle r \rangle^2\end{aligned}$$

$$\boxed{\langle \delta^2 \rangle = \langle r^2 \rangle - \langle r \rangle^2} \quad \left(\text{Dispersion of } r \right)$$

Dispersion: measures the amount of scatter of values of the variable about its mean value

The Width of the dispersion is given by

$$\boxed{\Delta r \equiv \sqrt{\langle (r - \langle r \rangle)^2 \rangle} = \sqrt{\langle r^2 \rangle - \langle r \rangle^2}}$$

$$\Delta r_2 = \sqrt{\langle r^2 \rangle_2 - \langle r \rangle_2^2}$$

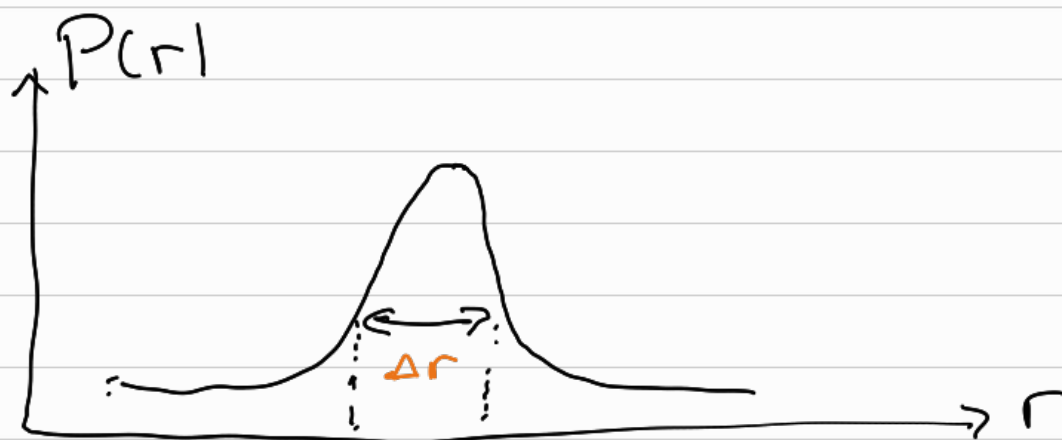
* Use this to solve the problem *

Note: you may have come across this in a probability course. It is also important in Statistical Mechanics.

- Also called the Root - Mean - Square (RMS) for obvious reasons

$$\Delta r = \sqrt{\langle (r - \langle r \rangle)^2 \rangle}$$

• Aside:



* Interpretation of Δr :

Δr is the "bulk width" of the probability distribution: (i.e. its where the particle is most likely to be). As such, we may understand Δr as the uncertainty in r .

- Here we looked at r but this is also true if we looked at the momentum $\langle p \rangle$ (or any other observable) to get

$$\Delta p = \sqrt{\langle (\hat{p} - \langle \hat{p} \rangle)^2 \rangle}$$

In fact, the (Generalized) Uncertainty Principle for two Hermitian operators $\hat{A}$ and $\hat{B}$ is:

$$\Delta \hat{A} \Delta \hat{B} \geq \frac{1}{2} \left| \frac{1}{i} \langle [\hat{A}, \hat{B}] \rangle \right|$$

ex: From elementary school we learned
 $[\hat{x}, \hat{p}] = i\hbar$

Thus:

$$\begin{aligned}\Delta \hat{x} \Delta \hat{p} &\geq \frac{1}{2} \left| \frac{1}{i} \langle [\hat{x}, \hat{p}] \rangle \right| \\ &\geq \frac{1}{2} \left| \frac{1}{i} \langle i\hbar \rangle \right| \\ &\geq \hbar/2\end{aligned}$$

or

$$\Delta \hat{x} \Delta \hat{p} \geq \frac{\hbar}{2}$$

- Relation we learn about early on, here we see how it emerges from our fancy maths.

